

MIT Art Design and Technology University's
MIT School of Computing, Pune
Department of Computer Science and Engineering
Class - TY ALL (SEM-I)
V sem Session 2024-25
OPERATING SYSTEM 21BTCS501

Assignment Questions – UNIT 1

1. Explain the primary functions of an operating system and discuss how it manages hardware resources effectively.
2. Describe the main components of a computer system and explain how they interact with each other to perform computing tasks.
3. Explain the differences between a monolithic kernel and a microkernel. Discuss the advantages and disadvantages of each approach.
4. Explain what a system call is and how it works. Provide examples of common system calls in UNIX-like operating systems and describe their purposes.
5. Identify the main categories of system calls. Describe how file management system calls differ from process control system calls, providing examples for each.
6. What are system programs, and how do they differ from application programs? Provide examples of common system programs and explain their roles in an operating system.
7. Discuss the key considerations in designing an operating system. How do operating systems achieve portability and compatibility across different hardware platforms?
8. Describe the concept of virtual machines. Compare type 1 and type 2 hypervisors, explaining their roles in virtual machine architecture and their respective advantages and disadvantages.
9. How do system programs differ from application programs?
10. What are some examples of common system programs?
11. How do layered operating systems improve system organization and reliability?
12. Explain Computer-System Organization with suitable diagram.
13. Define OS and Explain Functions of Operating Systems.
14. Enlist and explain different types of Application software.
15. Draw neat diagram and explain in detail computer system Architecture
16. What are the key differences between a monolithic kernel and a microkernel?
17. Enlist and explain different types of system software.
18. Enlist and explain services provided by operating system.
19. Explain with suitable diagram the operating system layered structure.
20. What are the different types of operating system structures? Explain any one operating system structure.